THE SYMPTOM LEDGER

Phase 1 submission draft for the NIH NOURISH: Autoimmunity Digital Health Challenge. $650,000 total purse; Phase 1 pays up to 10 prizes of $20,000; submissions due January 8, 2027. Judged on: understanding of the problem, solution scope, scientific rationale, engagement strategy, impact, feasibility, innovation. Each judgment criterion addressed in its own section below, mapped one-to-one. Her hand submits.

UNDERSTANDING OF THE PROBLEM

Autoimmune disease is a data problem wearing a clinical costume. Patients live with fluctuating, multi-system symptoms that average 3-5 years and multiple clinicians to diagnose; the signal that would shorten that path - symptom patterns over time, correlated with food, cycle, sleep, stress, and environment - is scattered across memory, notes apps, and the two minutes a clinic visit allows. flares get remembered as stories, not sequences. The unmet need is not another dashboard; it is a way for a person's lived state to become structured, clinician-readable evidence without turning the patient into a data-entry clerk.

THE SOLUTION: SCOPE

THE SYMPTOM LEDGER: a mobile-first digital health tool where a person presses their state into TEXT - free-form, in their own words, seconds per entry - and the tool deterministically structures it: symptom taxonomy mapping, temporal pattern extraction, trigger correlation, and a clinician-facing one-page timeline generated from the ledger. Phase 1 scope: the capture loop, the pattern engine, and the clinician summary, validated with a patient advisory group. Phase 2-3 scope: wearable ingestion and community cohort patterns.

SCIENTIFIC RATIONALE

Symptom diaries have clinical precedent; the failure mode is adherence and structure, both solvable. Free-text capture with deterministic (non-generative) structuring removes the hallucination class from a health context - every structured element traces to the person's own words, provably. Pattern extraction across cycles, meals, and sleep uses transparent, auditable methods - correlation surfaces the hypothesis, the clinician tests it. This is the no-black-box design: the tool shows its derivation, which is what makes it trustworthy enough to act on.

ENGAGEMENT STRATEGY

The instrument is built BY the population it serves: the principal architect's information architecture - pressing lived state into text - is the method, and the patient advisory group shapes the taxonomy from entry one. Engagement mechanics: entry takes under 30 seconds, the ledger answers back with the person's OWN patterns (not generic advice), and the clinician visit becomes the payoff moment - the tool hands the patient evidence that makes the appointment work.

IMPACT

Shorter time-to-diagnosis, better flare management, and a data asset autoimmunity research has never had: longitudinal first-person symptom structure at scale. Every structured ledger is consented, de-identified cohort data for the research community - the Phase 3 vision.

FEASIBILITY

Buildable by a two-principal team on a fixed clock: the structuring engine is deterministic software (no model training dependency), the capture UI is conventional mobile development, and the advisory-group protocol is standard patient-engagement methodology. Phase 1 lands well inside the January 8, 2027 window because nothing in the architecture waits on a research breakthrough - only on the build we run every day.

INNOVATION

The field's tools are either generative (untrustworthy in health) or forms (unused by week two). The unclaimed middle - deterministic structuring of free human text as a clinical instrument - is the innovation, and it descends from a working formal language (DELTA, 82/82 tests) built for exactly this: state in words, verified structure out.

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